

Department of Computer Science and Engineering
Indian Institute of Technology Jodhpur
CSL7490: Introduction to Blockchain (Quiz - 2)

Time: 20 Mins Date: 24 March 2026, Tuesday Maximum Marks: 20

Roll Number _____

Name: _____

Please read the following instructions carefully.

- All questions are compulsory. Each question carries one mark.
- Students must write their Name and Roll Number clearly on the question paper.
- Each question must be answered in exactly one sentence only, written directly on the question paper.
- No separate answer sheet shall be provided or accepted under any circumstances.
- Wherever required, students must clearly state and justify any assumptions made in their answers.
- Any form of copying or unfair means will result in zero marks in this quiz and may lead to further disciplinary action, including reduction of final course grades, as per the institute's rules.

1. Bitcoin's Script is Turing-incomplete by design, while the EVM is quasi-Turing complete. What is the primary security risk Bitcoin avoids by being Turing-incomplete?
Answer: The Halting Problem (infinite loops that could crash or hang all nodes).
2. During a transaction, the EVM uses three areas to store data: Stack, Memory, and Storage. Which of the following is true regarding their persistence and cost?
Answer: Storage is the only persistent area and is significantly more expensive than Memory or Stack.
3. When using the standard CREATE opcode, what two pieces of data are RLP-encoded and hashed to determine the address of a new Smart Contract?
Answer: The sender's address and the sender's current nonce
4. Why is it impossible for a Smart Contract Address to initiate a transaction on its own?
Answer: Because Contract Accounts do not have a corresponding private key; their actions are strictly governed by their internal bytecode triggered by an EOA.
5. If an EOA has a Nonce of n , and it sends a transaction that fails (reverts), what is the Nonce of that EOA after the block is finalized?
Answer: $n + 1$ (The nonce increments regardless of success or failure to prevent replay attacks)
6. If an Execution Layer transaction was modified to use BLS instead of ECDSA, what would be the primary disadvantage for a single, high-frequency trader?
Answer: BLS signature verification (Pairing) is computationally much more expensive for a single signature than ECDSA verification.
7. After The Merge, the Ethereum node was split into the Execution Client and the Consensus Client. How do these two layers synchronize to ensure a transaction is valid?
Answer: The Consensus Layer instructs the Execution Layer to "execute" a bundle of transactions via the **Engine API** and returns a **VALID** or **INVALID** status. or **Engine API**
8. Which of the following statements accurately describes the Address of a contract during the execution of its `init` code using `ADDRESS` opcode?
 - (a) The address does not exist until the `init` code finishes successfully.
 - (b) The address is already determined, and the `init` code can access it.

- (c) The address is temporary and changes once the `init` code is discarded.
- (d) The address is only valid if the `init` code contains a `DELEGATECALL`.

Answer: B

Analysis: The account is created before the `init` code runs, meaning the `init` code executes inside the context of the new address.

9. If a constructor sets a state variable (e.g., `uint public x = 10;`), where is that value 10 physically stored after the `init` code finishes?

Answer: In the account's permanent Storage trie.

10. Can you interact with a Smart Contract if you have the Address and the Bytecode, but you have LOST the JSON ABI? (Yes/ No and why?)

Answer: Yes, but you must manually Decompile the bytecode to figure out the 4-byte function selectors and their expected input types.

11. Contract A performs a `DELEGATECALL` to Contract B to execute a function that updates a user's balance. During this execution, Contract B's code runs a `SSTORE` instruction. Which of the following correctly describes the Contextual Identity of this operation?

- (a) `msg.sender` is Contract A, and the balance is updated in Contract B's storage.
- (b) `msg.sender` is the original Caller (EOA), and the balance is updated in Contract B's storage.
- (c) `msg.sender` is the original Caller (EOA), and the balance is updated in Contract A's storage.
- (d) `msg.sender` is Contract A, and the balance is updated in Contract A's storage.

Answer: C

12. Which of the following describes the specific function of an *Extension Node* in the Ethereum Merkle Patricia Trie (MPT)?

- (a) It contains a 17-entry array used to direct traversal across hexadecimal branches and a value field.
- (b) It is used to store the Keccak256 cryptographic hash of Ethereum block header data for verification purposes.
- (c) It holds the actual account-related data, such as balance, nonce, or contract storage value.
- (d) It compresses the path by storing a shared prefix and linking to the next Branch node.

Answer: D

13. What happens to the `v` value in an EIP-155 transaction compared to a legacy Ethereum transaction?

- (a) It includes the Chain ID to prevent replay attacks across different Ethereum-based networks.
- (b) It is replaced by a separate 48-byte BLS signature field used for validator authentication.
- (c) It is repurposed to record the priority fee or tip paid to the block proposer.
- (d) It is completely removed from the transaction format to reduce storage overhead in the block.

Answer: A

14. In the Ethereum Virtual Machine, a stack overflow occurs if the stack grows beyond _____ items.

Answer: 1024

15. A smart contract writes a value to memory and then the transaction finishes. What happens to that value?

- (a) It becomes permanent contract state.
- (b) It is stored in the block header.
- (c) It disappears after execution ends.
- (d) It is moved automatically into storage.

Answer: C

16. A consensus rule requires at least Supermajority Threshold support. If a committee has 150 validators, what is the minimum number needed to reach the threshold?

Answer: $(2/3) \times 150 = 100$

17. If two malicious block proposers can repeatedly cause honest validators to disagree on the head, which property is being threatened most directly?

Answer: Liveness can be threatened if disagreement is sustained and finalization is delayed.

18. What is the conceptual difference between the world state and storage?

Answer: World state is the global mapping of all accounts, while storage is the internal persistent data of a contract account.

19. When a contract is called, the EVM uses the _____ to look up the actual bytecode in the local database to load it into the virtual machine for execution.

Answer: codeHash

20. If there are 1,000,000 active validators, the protocol organizes them into 32 committees per slot. To ensure redundancy and efficiency, the protocol targets 16 Aggregators per committee. What is the probability that a single 32-ETH validator will be selected as an Aggregator for a specific slot?

Answer: 0.0512%

Step 1: Calculate total aggregators in a slot

$$\text{Committees per Slot} \times \text{Aggregators per Committee} = \text{Total Aggregator Slots}$$

$$32 \times 16 = 512 \text{ aggregator slots}$$

Step 2: Calculate individual probability P

$$P = \frac{\text{Total Aggregator Slots}}{\text{Total Active Validators}} = \frac{512}{1,000,000} = 0.000512$$

Step 3: Convert to percentage

$$0.000512 \times 100 = 0.0512\%$$